

Meniscal Allograft Transplant

ACG: A-0216 (AC)

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Clinical Indications for Procedure

- Current Role Remains Uncertain. Based on review of existing evidence, there are currently no clinical indications for this technology. See the Inconclusive or Non-Supportive Evidence section for more detailed analysis of the evidence base.(1)(2)(3)(4)(5)

Alternatives to Procedure

- Alternatives include(16):
 - Bracing
 - Meniscal repair or partial meniscectomy(17)
 - Pharmacotherapy (eg, anti-inflammatory medications)(18)
 - Physical therapy
 - Weight loss

Evidence Summary

Background

The medial and lateral knee menisci are semicircular wedge-shaped fibrocartilaginous structures that play a major role in shock absorption, load distribution, joint lubrication and congruity, and knee stabilization. Long-term meniscal deficiency has been shown to be a significant risk factor for the development of degenerative changes in the knee. The rationale for meniscal allograft transplant is to prevent the development or progression of early arthritis resulting from the loss of meniscal function in patients who have undergone early meniscectomy.(4)(5)(6)(7) **(EG 2)**

Inconclusive or Non-Supportive Evidence

The evidence for the recommendations found in this guideline includes 13 published peer reviewed articles and 2 book sections.

For prevention of osteoarthritis of the knee after meniscectomy, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review of 11 studies (658 patients with 688 procedures) evaluating the long-term survival and outcomes of meniscal allograft transplant found 10-year and 15-year allograft survival rates of 73.5% and 60.3%, respectively, and at a minimum of 10 years' follow-up, functional outcomes (measured by Lysholm, Tegner, Knee Injury and Osteoarthritis Outcome, or International Knee Documentation Committee (IKDC) scores) were improved compared with baseline. However, the authors noted that the low quality of evidence, small sample sizes, and heterogeneity of performed procedures and reported outcomes among the included studies limited the results.(8) **(EG 1)** A systematic review of 55

studies evaluating the efficacy of 1666 meniscal allograft transplants found, at a weighted average follow-up of 54 months, that there was improvement in function and pain scores, with a weighted failure rate of 8.7%. The authors stated that outcomes analysis for meniscal allograft transplant is problematic due to the preponderance of low-quality evidence, small sample sizes, varying surgical techniques, concomitant procedures, and lack of standardization of clinical outcome scores. Additional high-quality studies were recommended to understand the potential chondroprotective effect of the procedure.(1) **(EG 1)** A systematic review and meta-analysis of 24 studies (1882 knees) evaluating the outcomes of meniscal allograft transplant with and without other procedures (eg, autologous chondrocyte implantation, anterior or posterior cruciate ligament reconstruction, distal femoral or high tibial osteotomy) found, at a mean follow-up range of 25 to 180 months, that there was no significant difference between the groups with regard to pain, function, and activity scores. The authors noted that data were insufficient to determine the effects of osteotomy and cartilage procedures on the clinical outcomes of meniscal allograft transplant, and to compare complication, reoperation, graft survivorship, and graft failure rates between the groups.(9) **(EG 1)** A systematic review identified 44 studies (40 prospective cohort or case series and 4 retrospective case series) of 1136 meniscal allograft transplants, of which only 36% were performed as isolated procedures. Although the studies showed consistently high clinical success rates with an acceptable incidence of nonmajor complications, the authors acknowledged that the studies were of low methodological quality and higher-quality studies are needed.(10) **(EG 1)** A meta-analysis of 9 studies comparing medial (287) and lateral (407) meniscal allograft transplant survival found midterm (5 to 10 years) survival graft rates of 85.8% and 89.2%, respectively, and long-term (greater than 10 years) survival graft rates of 52.6% and 56.6%, respectively.(11) **(EG 1)** An observational study of 45 meniscal allograft transplants found, at mean follow-up of 8.6 years, that although there was significant improvement in pain and function scores, there was no significant improvement in activity scores. Of the entire cohort, 18% of transplants failed at a mean of 6 years, and 51% of patients required additional surgical treatment including chondral debridement or tear repair (13 patients), total knee replacement (4 patients), or meniscal excision (4 patients).(12) **(EG 2)** Observational studies of meniscal allograft transplants found, at mean follow-up of 15 to 16 years, that 24% to 28% of patients required total knee arthroplasty due to progression of osteoarthritis.(2)(13) **(EG 2)** A single-surgeon case series of 23 children (median age 17 years) who underwent meniscal allograft transplant found, at 7-year follow-up, that there was improvement in pain and function scores, as measured by IKDC score, Lysholm score, and Tegner activity score. The authors noted that the findings were limited by the sample size and heterogeneity of the study population.(14) **(EG 2)** An observational study of 36 adolescents who had undergone prior knee surgery found, at mean follow-up of 7 years, that meniscal allograft transplant was associated with improved pain, symptom, and function scores. However, the authors noted that because of the small sample size, it was impossible to correlate age, body mass index, previous surgery, and graft location with graft survival and clinical outcomes.(15) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment across patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 422.101(19); 42 CFR 419.22(20)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(21); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(22); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(23)

Medicare Coverage Determinations: Medicare Coverage Database(24)

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